

## Exponentials

$$e^{ix} = \cos x + i \sin x$$

$$\operatorname{Re}[e^{ix}] = \cos x$$

$$\operatorname{Im}[e^{ix}] = \sin x$$

$$[e^{ix}][e^{ix}]^* = \cos^2 x + \sin^2 x = 1$$

ck this:

$$e^{ix} e^{-ix} = 1 \checkmark$$

## Dirac delta function $\delta(x)$

A continuous distribution such that

$$\delta(x) = \begin{cases} \infty & x = 0 \\ 0 & x \neq 0 \end{cases}$$

and  $\int_{-\infty}^{\infty} \delta(x) dx = 1$

and  $\int_{-\infty}^{\infty} f(x) \delta(x) dx = f(0)$

and  $\delta(x-a) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iq(x-a)} dq$